

Routing Collapse in Classification-Head-Based Mixture-of-Experts Models for Aspect-Based Sentiment Analysis

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Abstract—Aspect-Based Sentiment Analysis (ABSA) requires identifying opinion targets and predicting the sentiment directed at each—a granularity that aggregate ratings cannot provide. We investigate whether a classification-head Mixture-of-Experts (MoE) architecture, knowledge distillation, and gradient-based token attribution can jointly improve accuracy, efficiency, and transparency on this task. Our central finding is that the MoE’s performance gain over a single-head BERT baseline stems not from expert specialization but from routing collapse: one expert dominates all predictions with a routing weight of ≈ 1.0 , making the model functionally equivalent to a deeper single MLP. This collapse also limits the benefit of subsequent knowledge distillation. These results underscore the importance of routing diagnostics and capacity-matched baselines when evaluating MoE systems in low-resource classification settings.

Index Terms—Aspect-Based Sentiment Analysis, Mixture of Experts, Routing Collapse, Knowledge Distillation, Token Attribution, Multi-Domain Sentiment Analysis

I. INTRODUCTION

Consider a laptop review that reads: “*The keyboard feels great, but the battery barely lasts four hours.*” A three-star aggregate rating reduces both opinions to a single number. Aspect-Based Sentiment Analysis (ABSA) keeps them separate: it identifies each aspect term (*keyboard*, *battery*) and classifies the sentiment directed at each as *positive*, *negative*, or *neutral* [2].

The business case for this granularity is compelling. A product manager who knows which features drive negative reviews can act on that signal in a way that a star-rating dashboard does not allow [13]. One practical barrier remains: deployed models rarely offer any explanation for their predictions, and a sentiment label without token-level attribution is difficult for a non-expert to interpret.

Many standard transformer ABSA baselines use one shared classification head across all inputs, a compromise when handling multiple domains. Task reformulation [3] and domain-adaptive pre-training [4] narrow this gap but do not address transparency.

From the perspective of information reuse, the four-stage framework introduced in Section III addresses both challenges. A shared encoder produces representations that are refined by

a Mixture-of-Experts (MoE) layer, whose learned outputs are transferred to a compact student, whose internal gradients are then used for token-level saliency attribution.

This article makes three contributions: (1) it proposes and evaluates a classification-head MoE architecture for combined-domain ABSA, including an analysis of expert routing behavior; (2) it shows that a compact distilled student retains most of the teacher’s performance at lower cost; and (3) it demonstrates that gradient saliency produces token-level attributions from the model’s own computation graph. Section IV evaluates each stage, and Section V discusses routing behavior and its implications.

II. RELATED WORK

The SemEval-2014 Task 4 benchmark [2] established the standard ABSA evaluation. Classical feature-based approaches [9], [13] often require domain-specific engineering or transfer poorly, motivating pre-trained transformers. Bidirectional Encoder Representations from Transformers (BERT) [1] enabled fine-tuned models for ABSA. BERT-SPC (Sentence Pair Classification) [3] and BERT-PT (Post-Training) [4] confirm fine-tuned transformers’ value for ABSA; their single-domain evaluations are not directly comparable to our combined-domain setup.

The Mixture-of-Experts (MoE) framework [5] replaces a single dense layer with independently parameterized expert networks, activating a subset per input via a learned gating function. Switch Transformer [6] adds load-balancing to discourage the concentration of all routing weight on a single expert. These works apply MoE inside transformer layers; we apply it at the classification-head level (Section IV-C).

Knowledge distillation [7] trains a compact student to match the teacher’s soft output distribution, preserving inter-class similarity beyond what hard labels convey. For token-level attribution, gradient norms [10] measure the local sensitivity of each token to the predicted logit, directly from the model’s computation graph. LIME (Local Interpretable Model-Agnostic Explanations) [11] and SHAP (SHapley Additive exPlanations) [12] are surrogate attribution methods also applied to

ABSA; we adopt gradient saliency because it operates directly on the computation graph without a surrogate.

III. FRAMEWORK

Our framework addresses three research questions: **RQ1**: Can a classification-head MoE improve multi-domain ABSA? **RQ2**: How much teacher knowledge can a compact student preserve? **RQ3**: Can the compact model’s internal signals yield human-readable rationales? We design a four-stage pipeline (Figure 1): an encoder baseline, an MoE teacher, a compact student via knowledge distillation, and a gradient saliency layer for symbolic attribution.

A. Notation

Let $x = (t_1, \dots, t_n)$ be an input sequence; subword tokenization may split a word into multiple tokens, so word-level labels require aggregating across constituent tokens. For ASC, the aspect is prepended: $x = (\text{aspect}, [\text{SEP}], t_1, \dots, t_n)$. An encoder maps x to $\mathbf{H} \in \mathbb{R}^{(n+2) \times d}$ (d : hidden dim.), yielding per-token vectors $\mathbf{h}_i \in \mathbb{R}^d$ and the sentence-level $\mathbf{h}_{[\text{CLS}]} \in \mathbb{R}^d$ (the [CLS] output aggregating the full sequence). Stages 1–2 share this encoder; Stage 3 uses a compact student encoder (Section IV).

B. Stage 1: Encoder Baseline

ABSA comprises two complementary sub-tasks. Aspect Term Extraction (ATE) is a *token-level* task: each token receives a Beginning/Inside/Outside (BIO) label (B-ASP, I-ASP, or O). Aspect Sentiment Classification (ASC) is a *sentence-level* task: given a review and an aspect term, classify the sentiment as *Positive*, *Negative*, or *Neutral*. Both tasks share the encoder but use separate linear projection heads.

For ATE, a linear projection $W_{\text{ATE}} \in \mathbb{R}^{3 \times d}$ maps each token vector $\mathbf{h}_i$ to a BIO label distribution via $\text{softmax}(W_{\text{ATE}}\mathbf{h}_i + b_{\text{ATE}})$, where $b_{\text{ATE}} \in \mathbb{R}^3$ and $y_i \in \{O, B\text{-ASP}, I\text{-ASP}\}$. For ASC, an analogous projection maps $\mathbf{h}_{[\text{CLS}]}$ to a sentiment class. Both heads are optimized jointly with cross-entropy loss using AdamW.

C. Stage 2: Mixture-of-Experts Teacher

Stage 1’s ASC head applies identical parameters to all inputs regardless of domain. A model trained on both laptop and restaurant reviews must represent a compromise between two quite different vocabularies and phrasing conventions. We hypothesize (RQ1) that routing different inputs to independently trained expert networks can reduce this compromise without requiring explicit domain labels at inference time.

We replace the single ASC head with $K = 4$ independent expert networks $\{E_1, \dots, E_K\}$ controlled by a learned gating function, following standard practice for small-scale MoE experiments. For each input, a gating network selects the top $k = 2$ experts (top-2 routing). The shared encoder is fine-tuned jointly, also improving ATE performance (Table II).

Gating. A two-layer MLP $g_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^K$ maps the sentence-level representation $\mathbf{h}_{[\text{CLS}]}$ to a scalar score g_j for each expert $j \in \{1, \dots, K\}$, forming the active set $\mathcal{S}$ of size $k = 2$ with

softmax-normalized routing weights, where $w_j \in [0, 1]$ denotes the weight assigned to expert j :

$$\mathcal{S} = \text{top-2}(g_\phi(\mathbf{h}_{[\text{CLS}]})$$

$$w_j = \frac{\exp(g_j)}{\sum_{r \in \mathcal{S}} \exp(g_r)}, \quad j \in \mathcal{S} \quad (1)$$

By construction $\sum_{j \in \mathcal{S}} w_j = 1$. We route through $\mathbf{h}_{[\text{CLS}]}$ because it is the standard sentence-level representation used for classification; its suitability as a routing signal is examined empirically in Section IV-C.

Expert output. Each expert $E_k : \mathbb{R}^d \rightarrow \mathbb{R}^3$ is a three-layer MLP with GELU activations and hidden dimensions ($d \rightarrow 256 \rightarrow 128 \rightarrow 3$). The final logits are:

$$\hat{y} = \sum_{j \in \mathcal{S}} w_j \cdot E_j(\mathbf{h}_{[\text{CLS}]}) \quad (2)$$

Load-balancing. To discourage routing collapse, we add an auxiliary entropy term over batch-averaged weights $\bar{w}_k = \frac{1}{B} \sum_{b=1}^B w_k^{(b)}$:

$$\mathcal{L}_{\text{aux}} = - \sum_{k=1}^K \bar{w}_k \log \bar{w}_k \quad (3)$$

This is the routing entropy ($\log K \approx 1.39$ at uniform, 0 at full collapse); minimizing it inadvertently encourages collapse rather than diversity. Total loss: $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{aux}}$ ($\lambda = 0.1$).

D. Stage 3: Knowledge Distillation

Knowledge distillation (KD) [7] addresses RQ2 by training a compact student to match the teacher’s *soft labels*, which encode inter-class similarity beyond one-hot supervision.

Let $\mathbf{z}^T, \mathbf{z}^S \in \mathbb{R}^3$ be the teacher and student logits respectively, and $\sigma(\cdot)$ denote the softmax function. The Kullback–Leibler (KL) divergence between the temperature-scaled distributions forms the distillation objective:

$$\mathcal{L}_{\text{KD}} = \alpha T^2 \text{KL}\left(\sigma\left(\frac{\mathbf{z}^T}{T}\right) \parallel \sigma\left(\frac{\mathbf{z}^S}{T}\right)\right) + (1 - \alpha) \mathcal{L}_{\text{CE}}^S \quad (4)$$

where temperature $T > 1$ softens the teacher’s distribution to expose inter-class probability mass; T^2 compensates for the reduced gradient magnitude [7]. The weight $\alpha \in [0, 1]$ balances the soft-label term against the hard cross-entropy loss $\mathcal{L}_{\text{CE}}^S$; specific values are in Section IV.

E. Stage 4: Gradient Saliency Attribution

Even a well-performing compact model remains opaque: it assigns a label but offers no indication of *why*. We address this (RQ3) by computing gradient saliency—computing the ℓ_2 norm of the gradient of the predicted logit with respect to each token embedding measures the local sensitivity of the prediction to that token, directly from the student’s computation graph, without fitting a separate surrogate model.

Formally, the saliency score s_i for token t_i is:

$$s_i = \|\nabla_{\mathbf{e}_i} \hat{y}_{\hat{c}}\|_2 \quad (5)$$

where $\hat{c}$ is the predicted class. Because the ℓ_2 norm discards direction and sign, s_i reflects sensitivity rather than a causal contribution to the prediction. The top- m non-special tokens by s_i (with $m = 3$) form the attribution set $\mathcal{E} = \{e_1, \dots, e_m\}$, and the result is reported as `aspect → sentiment →  $\mathcal{E}$` .

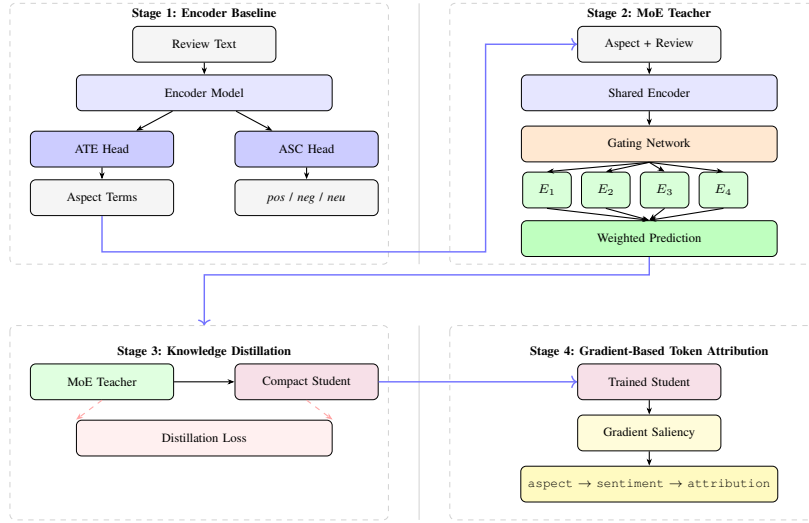


Fig. 1. Overview of the four-stage ABSA framework. Stage 1 establishes an encoder baseline with separate ATE and ASC heads. Stage 2 builds the MoE teacher (the shared encoder also trains an ATE head, not shown for clarity). Stage 3 trains a compact student via the combined distillation loss (Eq. (4)), which includes both soft- and hard-label terms. Stage 4 computes gradient saliency to produce token-level attributions.

IV. EXPERIMENTS AND RESULTS

A. Data and Setup

To evaluate our three research questions, we use the SemEval 2014 Task 4 benchmark [2], which provides annotated reviews from two distinct domains: Laptop reviews and Restaurant reviews. These two domains differ substantially in vocabulary—making them a natural test case for RQ1 (whether MoE routing can exploit domain structure). The two domains are combined while preserving the original SemEval train/test partitions. Table I gives the resulting statistics; all counts are *aspect-instance* counts (sentence–aspect pairs), since a sentence can contain multiple aspects. The combined test set contains 1,920 aspect instances for ATE evaluation; of these, 1,215 also carry unambiguous ASC labels and are used for the routing analysis and sentiment evaluation. Class proportions are moderately imbalanced (*Neutral* at 18%), so macro-F1 is the primary evaluation criterion.

TABLE I
DATASET STATISTICS (ASPECT-INSTANCE COUNTS)

Domain	Train	Test	Pos / Neg / Neu
Laptop	3,048	800	47% / 33% / 20%
Restaurant	3,813	1,120	59% / 23% / 18%
Combined	6,861	1,920	54% / 28% / 18%

All models use PyTorch 2.x with Hugging Face Transformers. Stages 1–2 use `bert-base-uncased` [1] (110M params, 12 layers, 768-d); Stage 3 uses `distilbert-base-uncased` [8] (66M params, 6 layers). Training: AdamW, batch 16–32, 10 epochs, max seqlen 128, primary seed 42, NVIDIA T4. Stage 2: $\lambda = 0.1$, $K = 4$, top- $k = 2$. Stage 3: $T = 4.0$, $\alpha = 0.7$. To assess sensitivity to random initialization, Stages 1 and 2 were additionally trained with seeds {0, 123}.

B. Performance

Table II compares accuracy and macro-F1 across the model stages. Because our setup combines domains, published single-domain results (e.g. BERT-SPC, BERT-PT) are not directly comparable and are omitted.

TABLE II
MODEL PERFORMANCE (% , MACRO-F1 IN BOLD)

Model	ATE Acc	ATE F1	ASC Acc	ASC F1
BERT-Base	96.94	90.50	81.32	77.66
MoE-Teacher	97.28	91.50	82.06	78.67
MLP-Head [†]	–	–	82.06	78.90
MoE-Student	–	–	80.16	75.51
Student-CE [‡]	–	–	–	78.44
Δ (MoE vs. Base)	+0.34	+1.00	+0.74	+1.01
F1 retention	–	–	–	96.0%

[†]Capacity-matched MLP ($d \rightarrow 256 \rightarrow 128 \rightarrow 3$), no routing.

[‡]DistilBERT trained with hard labels only (no teacher).

MoE-Teacher outperforms BERT-Base on every metric (seed 42); as discussed in Section IV-C, this gain is attributable to head capacity rather than routing specialization. MoE-Student retains 96.0% of teacher F1 (75.51 vs. 78.67), consistent with RQ2. A hard-label-only student (Student-CE) achieves 78.44%—confirming that a collapsed teacher’s soft labels add little inter-class structure beyond hard labels. Across seeds {42, 0, 123}, BERT-Base is $77.49 \pm 1.34\%$ and MoE-Teacher $78.16 \pm 1.48\%$, with overlapping ranges ($\Delta = +0.67\%$): the gain is within seed variance.

C. Routing Analysis

With top-2 routing, each example activates exactly two experts; activation rates therefore sum to 200% across all four experts. Table III reports per-expert activation rates and average weights on the ASC test set ($n = 1,215$).

TABLE III
EXPERT ROUTING STATISTICS ($n = 1,215$; TOP-2 ROUTING)

Expert	Active (%)	Avg. weight	Status
Expert 1	0%	—	Dormant
Expert 2	100%	0.9998	Dominant
Expert 3	64%	0.0002	Negligible
Expert 4	36%	0.0002	Negligible

Expert 2 is present in every example’s top-2 pair and carries a routing weight of ≈ 1.0 ; Expert 1 is never selected; Experts 3–4 receive negligible weight (≈ 0.0002). The auxiliary loss (3) proved insufficient to prevent this collapse. Since Expert 2 dominates all routing decisions, the effective model is a single expert MLP. To directly test whether head depth drove the gain, we trained a capacity-matched MLP head ($d \rightarrow 256 \rightarrow 128 \rightarrow 3$) without routing (MLP-Head). This achieves 78.90% ASC F1—indistinguishable from BERT-Base (77.66%) and MoE-Teacher (78.67%) within seed variance (± 1.34 – 1.48%). Neither head depth nor routing produces a statistically significant gain at this scale.

λ ablation. To investigate why the auxiliary loss failed, we trained MoE with $\lambda \in \{0.01, 0.10, 0.50, 1.00\}$. Increasing λ worsens collapse (dominant weight: $0.62 \rightarrow 0.9999$), confirming that minimizing $\mathcal{L}_{\text{aux}}$ collapses routing entropy to zero rather than maximizing it—inadvertently encouraging collapse. A corrected implementation (e.g. maximizing entropy via a sign-flipped loss) is left for future work.

D. Symbolic Rationales

Table IV shows four illustrative token attributions produced by Stage 4 via (5).

TABLE IV
SAMPLE SYMBOLIC RATIONALES FROM STAGE 4

Domain	Aspect	Sentiment	Top Tokens
Restaurant	price	<i>Negative</i>	not, nearly, warrant
Laptop	battery life	<i>Positive</i>	long, very, best
Laptop	price	<i>Positive</i>	competitive, low
Restaurant	menu	<i>Neutral</i>	tried, their

Attribution scores are most interpretable for *Positive* and *Negative* examples; *Neutral* examples yield weaker and less consistent signals, reflecting their inherently ambiguous sentiment.

V. DISCUSSION

The routing collapse observed in our experiments suggests that pooled sentence representations may be insufficient routing signals for classification-head MoE systems. Richer signals—per-token attention weights or explicit domain embeddings—may be required before routing diversity materializes in this setting. The capacity gain from a deeper expert MLP, however, appears real under this configuration.

Limitations. All reported differences are within seed variance (± 1.34 – 1.48%); the MoE gain ($\Delta = +0.67\%$) is not statistically significant at this scale. A capacity-matched MLP head (MLP-Head) and a hard-label-only student (Student-CE) confirm that neither head depth nor soft-label distillation

produces a meaningful gain when routing collapses to a single expert. Routing collapse itself stems from an entropy-minimizing auxiliary loss that inadvertently encourages collapse (Section IV-C). Rationale quality (Table IV) is qualitative only; no faithfulness test was conducted.

VI. CONCLUSION

We presented a four-stage ABSA framework: an encoder baseline, a Mixture-of-Experts (MoE) teacher, a compact distilled student, and a gradient-saliency rationale layer. MoE gains arise from expert capacity rather than routing specialization—routing collapsed to a single dominant expert despite load-balancing. Knowledge distillation preserved inter-class structure at lower parameter cost, and gradient saliency produced token-level attributions from the model’s computation graph. Future work should adopt per-token routing signals and a corrected load-balancing loss.

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